BEST BRAIN EXAMINATIONS KONSORTIUM

SPECIAL MOCK EXAMINATIONS FOR BECE CANDIDATES – OCTOBER 2026

MARKING SCHEME – MATHEMATICS

PAPER 2 [60 MARKS]

GENERAL NOTES ON PAPER II

1. Marks are subdivided into marks for method (M), marks for accuracy (A) and marks for accuracy not preceded by M marks (B).
2. The M marks should be given for a particular stage if the method is correct, that is, it would yield the right answer, if correctly carried out without numerical errors. M marks are not generally subdivided and unless the M mark for a preceding stage has been awarded, no A marks can be gained for that stage. (No deduction should be made from M).
3. Deduct 1 mark for omission of units or for wrong units not more than once in one whole question.
4. Give '0' (zero) for results obtained for work that is indecipherable or wholly suppressed.
5. If more questions are attempted than the rubric allows, delete the marks given for these extra questions which have the lowest marks. This rule implies that for candidates attempting more than four questions, consider only their best four and ignore the remaining questions by writing "MQA". (MQA denotes More Questions Answered than allowed)
6. Do not mark beyond the first appearance of a correct answer, i.e. ignore any further work beyond the correct answer.

No.	SOLUTION	MARKS
1.	**QUESTION ONE**	
(a)(i)	Expressing 31 days in minutes 60 minutes = 1 hour; 1 day = 24 hours Number of minutes in a day = 24 hrs × 60 mins = 1440 mins In 31 days = 31 × 1440 mins = 44 640 minutes In standard form = 4.4640×10^4	M1 *(minutes in a day)* M1 A1
(ii)	6 hours to seconds 3600 secs = 1 hr 6 hours = 6 × 3600 s = 21 600 s Standard form = 2.16×10^4	M1 A1
(b)	**Fred** Monthly earnings = GH¢2 500; 5 years = 5 × 12 = 60 months Five years' earnings = GH¢2 500 × 60 = GH¢150 000 **Eben** Monthly earnings = 3 × GH¢2 500 = GH¢7 500 Five years' earnings = GH¢7 500 × 60 = GH¢450 000 ∴ Total = GH¢150 000 + GH¢450 000 = GH¢600 000	M2 A1
(c)	**Value of m** 2, 3, (4m − 3), (3m + 1), 11, 13 [(4m − 3) + (3m + 1)] ÷ 2 = 6 (7m − 2)/2 = 6 ⇒ 7m − 2 = 12 7m = 14 ⇒ m = 2	M1 *(median equation)* M1 (solving) A1 (m = 2)

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No.	SOLUTION	MARKS
(d)(i)	**Stem and Leaf Plot** Stem 3 | 8 Stem 4 | 0, 2, 5, 8 Stem 5 | 4, 4, 6, 7, 7, 7, 8, 8 Stem 6 | 1, 3, 3, 6, 8 Stem 7 | 2, 3	B2 $\frac{1}{2}$ −EE) *(data correctly placed in stems)*
(ii)	Mode = 57	B$\frac{1}{2}$ *(mode = 57)*
(ii)	Arranged: 38, 40, 42, 45, 48, 54, 54, 56, 57, 57, 57, 58, 58, 61, 63, 63, 66, 68, 72, 73 Median = (57 + 57)/2 = 114/2 Median = 57	M$\frac{1}{2}$ *(middle position)* A1 *(median = 57)*
		QUESTION 1 TOTAL = 15 MARKS
2.	**QUESTION TWO**	
(a)(i)	Quantity Amina had = 2½ litres; Quantity drank = ⅓ litre Quantity left = 2½ ℓ − ⅓ ℓ = 5/2 ℓ − 1/3 ℓ = (15ℓ − 2ℓ)/6 = 13ℓ/6 Quantity of water left = 2⅙ litres	M1 A1 (2⅙ litres)
(ii)	0.893 × 14.7 = 893/1000 × 147/10 = (893 × 147)/(1000 × 10) = 131 271/10 000 = 13.1271	M1 *(converting to fractions)* A1
(b)	Mean height of 28 students = 162 cm Mean = Total height ÷ Number of students 162 = Total height ÷ 28 ⇒ Total height = 162 × 28 = 4 536 New total = 4 536 + 149 = 4 685 New mean = 4 685 ÷ 29 = 161.55 cm	M1 *(total height = 4 536)* M1 A1
(c)(i)	$(-3)^{(x-2)} = -243$ $(-3)^{(x-2)} = (-3)^5$ $x - 2 = 5 \Rightarrow x = 7$	M1 A1 *(x = 7)*
(ii)	Water consumption = $11^{(4x-3)}$, x in months; when x = 2 $= 11^{(4(2)-3)} = 11^{(8-3)} = 11^5$ = 11 × 11 × 11 × 11 × 11 = 161 051 Total consumption = 161 100 (4 significant figures)	M1 M$\frac{1}{2}$ *(substituting x = 2)* A$\frac{1}{2}$ *(161 100 to 4 s.f)*
(d)(i)	**Frequency table** x = 1, f = 4, fx = 4 x = 2, f = 5, fx = 10 x = 3, f = 42 ÷ 3 = 14, fx = 42 x = 4, f = 9, fx = 36 x = 5, f = 30 ÷ 5 = 6, fx = 30 x = 6, f = 2, fx = 12	B2 *(completed table − $\frac{1}{2}$EE)*

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No.	SOLUTION	MARKS
	$4 + 9 = 21x \Rightarrow 13 = 21x$ $13/21 = 21x/21 \Rightarrow x = 13/21$	 A1
(c)	$(25 \times t^{-4}) / (5^{-3} \times 10 \times t^{-8})$ $= (5^2 \times t^{-4}) / (5^{-3} \times (5 \times 2) \times t^{-8})$ $= 5^2/(5^{-3} \times 5^1) \times ½ \times t^{-4}/t^{-8}$ $= 5^2/5^{-2} \times ½ \times t^{-4}/t^{-8}$ $= 5^{\wedge}(2 - (-2)) \times t^{\wedge}(-4 - (-8)) \times ½$ $= (5^4 \times t^4)/2$ $= (625 \times t^4)/2$	M1 M1 M1 A1
(d)	$a = (-4, p)$ and $b = (q, 1)$ [column vectors] Then $a + b = (-1, -8)$ $-4 + q = -1 \Rightarrow q = -1 + 4 \Rightarrow q = 3$ $p + 1 = -8 \Rightarrow p = -8 - 1 \Rightarrow p = -9$ $\therefore p = -9, q = 3$	M1 M$\frac{1}{2}$ M$\frac{1}{2}$ A1
		QUESTION 6 TOTAL = 15 MARKS

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PAPER 1 [40 MARKS]

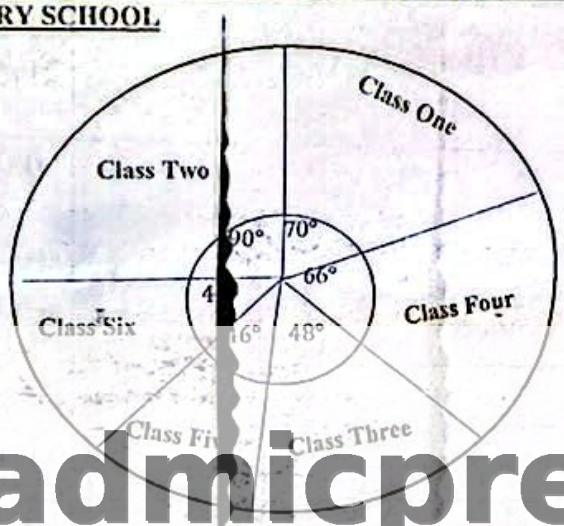
1. A	11. B	21. D	31. A
2. B	12. C	22. C	32. C
3. C	13. A	23. B	33. C
4. A	14. C	24. A	34. B
5. B	15. D	25. C	35. A
6. C	16. A	26. C	36. D
7. C	17. A	27. D	37. B
8. C	18. A	28. D	38. C
9. B	19. C	29. B	39. C
10. B	20. D	30. B	40. C

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No.	SOLUTION	MARKS
	$\Sigma f = 40,\ \Sigma fx = 134$	
(ii)	Students in the class = 4 + 5 + 14 + 9 + 6 + 2 = 40	B$\frac{1}{2}$ *(40 students)*
(iii)	Marbles brought altogether = 4 + 10 + 42 + 36 + 30 + 12 = 134	B1 *(134 marbles)*
(iv)	Most of the students brought 6 marbles	B$\frac{1}{2}$ *(3 marbles)*
		QUESTION 2 TOTAL = 15 MARKS
3.	**QUESTION THREE**	
(a)	Wage = GH¢8¼ per hour **Overtime rate = 1½ × 8¼ = 3/2 × 33/4 = GH¢99/8 per hour** Amount Mark earns in a day = (8¼ × 8) + (99/8 × 4) = (33/4 × 8) + (99/8 × 4) = (33 × 2) + (99/2) = 66 + 49.5 = 115.5 ∴ Mark earns GH¢115.50 in a day	M1 M1 M1 A1 *(GH¢115.50)*
(b)(i)	From Pythagoras' theorem: $H^2 = O^2 + A^2$ $\|PQ\|^2 = \|PR\|^2 + \|QR\|^2$ $\|PQ\|^2 = (8\text{ cm})^2 + (6\text{ cm})^2 = 64\text{ cm}^2 + 36\text{ cm}^2$ $\|PQ\|^2 = 100\text{ cm}^2$ $\|PQ\| = \sqrt{100} = 10\text{ cm}$	M1 *(Stating Pythagoras theorem)* M1 ($\|PQ\|^2 = 100$) A1 ($\|PQ\| = 10$ cm)
(ii)	tan θ = Opposite / Adjacent tan θ = 8 cm / 6 cm tan θ = 4/3	M1 *(correct ratio)* A1 (4/3)
(c)	$6 \geq 2 - 3x$ $3x \geq 2 - 6 \Rightarrow 3x \geq -4$ $3x/3 \geq -4/3 \Rightarrow x \geq -4/3$ or $-1\frac{1}{3}$ Truth set: $\{x : x \geq -4/3\}$ Representation on a number line -3 -2 -1⅓ -1 0 1 2 3 4	M1 *(grouping terms)* M1 *(dividing by 3)* A1 ($x \geq -4/3$) B1 $-\frac{1}{2}$EE *(number line*
(d)	Given radius = 7 cm Area of a semi-circle = ½ × πr² = ½ × 22/7 × 7 cm × 7 cm = 11 × 1 cm × 7 cm = 77 cm²	M1 *(correct formula)* A1 (77 cm²)
		QUESTION 3 TOTAL = 15 MARKS

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No.	SOLUTION	MARKS
	QUESTION FOUR	
(a)	Total number of pupils = 35 + 45 + 24 + 33 + 23 + 20 = 180 **Angles of sectors:** One: 35/180 × 360° = 70° Two: 45 × 2° = 90° Three: 24 × 2° = 48° Four: 33 × 2° = 66° Five: 23 × 2° = 46° Six: 20 × 2° = 40° Total = 180 pupils; 360°	*B2½ (− ½EE)* *all angles correct*
(i)	**A PIE CHART ILLUSTRATING THE NUMBER OF PUPILS IN PRIMARY SCHOOL**  	B2½ (½EE) *drawing & labelling pie chart −*
(ii)	Fraction of pupils in primary 4 45/180 = 9/36 = 1/4 OR 90°/360° = 1/4	A½
(iii)	Percentage of pupils in Primary three, four and five Pupils in Primary 3, 4 and 5 = 24 + 33 + 23 = 80 Percentage = 80/180 × 100% = 8/18 × 100% = 4/9 × 100% = 400/9 % = 44.444% Percentage = 44.44% (2 decimal places)	M1 A1
(b)(i)	Gradient (m) = 7; Point P(−2, −3) Equation of the line $y - y_1 = m(x - x_1)$ $y - (-3) = 7(x - (-2))$ $y + 3 = 7(x + 2)$ $y + 3 = 7x + 14$ $y = 7x + 14 - 3 \Rightarrow y = 7x + 11$ OR $y - 7x - 11 = 0$	M½ M½ A1

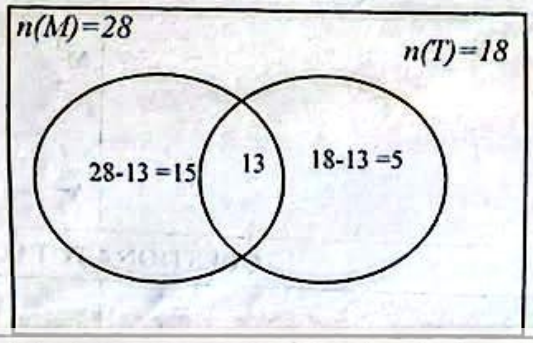
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No.	SOLUTION	MARKS
(ii)	The value of k; R(k, 4) lies on $y = 7x + 11$ When $x = k$ and $y = 4$: $4 = 7k + 11$ $4 - 11 = 7k \Rightarrow -7/7 = 7k/7$ $-1 = k$ ∴ The value of k is -1	M$\frac{1}{2}$ A1
(c)	$2/3 \times 1/4 + 2/5 \div 14/15$ $= (2/3 \times 1/4) + (2/5 \div 14/15)$ $= 2/12 + 3/7$ $= 1/6 + 3/7$ $= 25/42$	M1 A1
(d)	Length = (x + 2), Breadth = (2x + 1), Perimeter = 48 km Perimeter = 2L + 2B ⇒ 2L + 2B = 48 $2(x + 2) + 2(2x + 1) = 48$ $2x + 4 + 4x + 2 = 48$ $6x + 6 = 48 \Rightarrow 6x = 42$ $6x/6 = 42/6 \Rightarrow x = 7$ km	M1 M$\frac{1}{2}$ A$\frac{1}{2}$
		QUESTION 4 TOTAL = 15 MARKS
5.	QUESTION FIVE	
(a)(i)	Given Number of red pens, $n(red) = 5$, Number of blue pens $n(blue) = 3$ and Number of green pens, $n(green) = 2$ Total number of pens = $n(red) + n(blue) + n(green)$ $= 5 + 3 + 2 = 10$ Probability of picking a red pen, $P(red)$ $= \frac{n(red)}{Total\ number\ of\ balls}$ $= \frac{5}{10}$ $= \frac{1}{2}$	B$\frac{1}{2}$ M1 A$\frac{1}{2}$
(ii)	$P(\text{blue}) = \frac{n(blue)}{Total\ number\ of\ balls} = \frac{3}{10}$ (α) P(blue) as a decimal = 0.3 (β) P(blue) as a percentage = $\frac{3}{10} \times 100 = 30\%$ (γ) P(blue) as a ratio = 3:10	B$\frac{1}{2}$ B$\frac{1}{2}$ B$\frac{1}{2}$ B$\frac{1}{2}$
(iii)	Probability of picking a first green = $\frac{n(green)}{Total\ number\ of\ balls}$ $= \frac{2}{10} = \frac{1}{5}$ Probability of picking a 2nd green = $\frac{n(green)-1}{Total\ number\ of\ balls-1} = \frac{1}{9}$ Probability that both marbles are green = $\frac{1}{5} \times \frac{1}{9} = \frac{1}{45}$	M1 M1 M$\frac{1}{2}$A$\frac{1}{2}$

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No.	SOLUTION	MARKS
(b)	Departure Time = 11:13 am; Arrival Time = 3:08 pm = 15:08 GMT 15 : 08 − 11 : 13 = 3 : 55 The journey took 3 hours 55 minutes	M1 A$\frac{1}{2}$
(c)	Number of gallons delivered per minute = 20 Duration Kofi can afford = ½ hr = 30 minutes Number of gallons over the 30-minute period = 30 × 20 = 600 gallons	M1 A$\frac{1}{2}$
(d) (i)	Let M = the set of fishmongers who sell mackerel; T= the set of fishmongers who sell tilapia $n(M) = 28, n(T) = 18, n(M \cap T) = 13$  	B3 − $\frac{1}{2}$EE
(ii)	Let Number of fishmongers who do not sell any of the two = x $\Rightarrow 15 + 13 + 5 + x$ $= 36 \Rightarrow 33 + x$ $= 36 \Rightarrow x$ $= 36 - 33 = 3$ ∴ 3 students are not involved in any activity	M1 A1
		QUESTION 5 TOTAL = 15 MARKS
6.	QUESTION SIX	
(a)	Let A_l – area of the larger rectangle, A_s = area of the smaller rectangle **Area of the shaded region = $A_l - A_s$** **Area of a rectangle = Length × Width** $A_l = (5x + 8)(6x + 2)$ $= 30x^2 + 10x + 48x + 16$ $A_l = 30x^2 + 58x + 16$ $A_s = 5x(x + 6) = 5x^2 + 30x$ Area of shaded region = $30x^2 + 58x + 16 - (5x^2 + 30x)$ $= 30x^2 + 58x + 16 - 5x^2 - 30x$ $= 25x^2 + 28x + 16$	M1 M1 M1 A1
(b)	$4/(3x) = 7 - 3/x$ Multiply through by the L.C.M. = 3x $(3x)\cdot 4/(3x) = (3x)\cdot 7 - (3x)\cdot 3/x$ $4 = 21x - 9$	M1 M1 M1

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